React Application — Production Deployment

DevOps Project Submission Document | Author: Sudirpandian | Date: June 2026

GitHub Repository: https://github.com/sudirpandian/Devops/tree/project-2 | **Branch:** project-2

1. Project Overview

This document covers the end-to-end production deployment of the Trend React application, including containerization, infrastructure provisioning, Kubernetes deployment, CI/CD pipeline, and monitoring.

Component	Technology	Details
Application	React (Vite build)	Served via Nginx on port 3000
Container	Docker	sudirpandian01/trend-app
Infrastructure	Terraform	AWS VPC, EC2, Security Group, IAM
Kubernetes	AWS EKS	trend-eks, ap-south-1, 2 nodes
CI/CD	Jenkins	Declarative pipeline with GitHub webhook
Monitoring	Prometheus + Grafana	kube-prometheus-stack via Helm

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2. Application Setup

2.1 Clone Repository

```
git clone https://github.com/Vennilavanguvi/Trend.git
cd Trend
ls -la
```

```
[root@ip-172-31-11-132 project]# tree Trend/
Trend/
├── README.md
└── dist
    ├── assets
    │   ├── About-DbHf8GFx.js
    │   ├── Cart-CKKAW61b.js
    │   ├── CartTotal-4jDB2az0.js
    │   ├── Collection-CSvJySr8.js
    │   ├── Contact-BomtQVpG.js
    │   ├── ForgotPassword-B3rsrKs1.js
    │   ├── Home-Bj1xAQU1.js
    │   ├── Login-B3PL5s7p.js
    │   ├── NewsLetterBox-h8A4PbZy.js
    │   ├── Orders-CaCJbx18.js
    │   ├── PlaceOrder-D9bDxOFB.js
    │   ├── Product-gEYDQbwF.js
    │   ├── ProductItem-CABqN4-8.js
    │   ├── Profile-C4iJY7-d.js
    │   ├── ResetPassword-Dp6S-2CI.js
    │   ├── Title-D4NmpLxE.js
    │   ├── WishList-DSfx7B72.js
    │   ├── about_img-BAJyTXw9.png
    │   ├── contact_img-CyOum2vk.png
    │   ├── hero_img-DOCOb6wn.png
    │   ├── index-B52p61-W.css
    │   ├── index-BjbOuBLy.js
    │   ├── logo-BDGV9g83.png
    │   ├── p_img1-BTuXsnJl.png
    │   ├── p_img10-Bw8q3bg_.png
    │   ├── p_img11-BSX5s8O8.png
    │   ├── p_img12-CWT58dgm.png
    │   ├── p_img13-9keE2fyF.png
    │   ├── p_img14-3-oykRil.png
    │   ├── p_img15-CEqmdW8L.png
    │   ├── p_img16-DpIAWCPe.png
    │   ├── p_img17-D3TjESyL.png
    │   ├── p_img18-CZUsXTNh.png
    │   ├── p_img19-Ddd8N1Sm.png
    │   ├── p_img20-BN74rvum.png
    │   ├── p_img21-BhDxXaVk.png
    │   ├── p_img22-DIBExxK1.png
    │   ├── p_img23-CGTC-19z.png
    │   ├── p_img24-DxfyHMhA.png
    │   ├── p_img25-DpcquDQp.png
    │   ├── p_img26-DAHi2sy3.png
    │   ├── p_img28-CJDN8kuL.png
    │   ├── p_img2_1-ku4Fyxmj.png
```

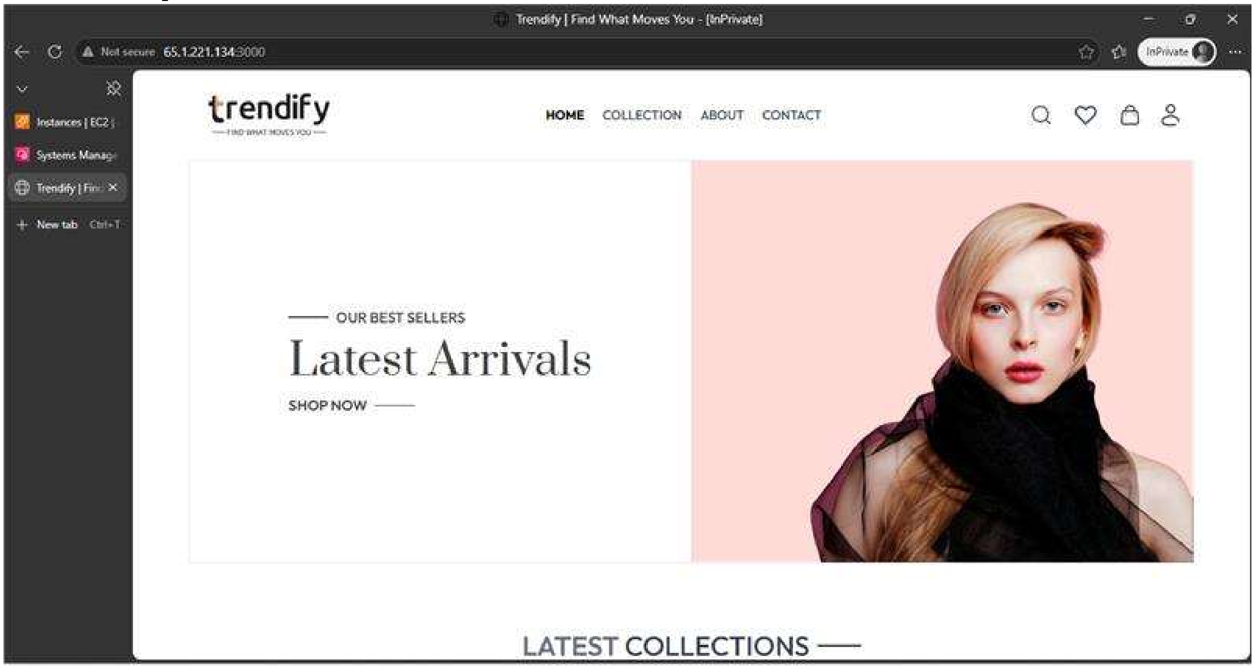
```
    │   ├── p_img48-DwPRHR8_.png
    │   ├── p_img49-CghJrhGi.png
    │   ├── p_img5-BMHJXi1_.png
    │   ├── p_img50-Cg-7CHi5.png
    │   ├── p_img51-FRQODGKA.png
    │   ├── p_img52-CY-YbGLu.png
    │   ├── p_img6-B2VXR-Iw.png
    │   ├── p_img7-Ds_MiPCY.png
    │   ├── p_img8-BeUACWv0.png
    │   ├── p_img9-DmR0Ahyl.png
    │   └── razorpay_logo-DrY6yMWi.png
    ├── index.html
    └── vite.svg

 directories, 74 files
root@ip-172-31-11-132 project]# cd Trend/
```

The repository contains a pre-built Vite/React application with only a `dist/` folder — no Node.js source. Static files are served directly via Nginx.

2.2 Run Application Locally (Port 3000)

```
docker run -d -p 3000:3000 trend-app:v1
curl -I http://localhost:3000
```



3. Docker

3.1 Dockerfile

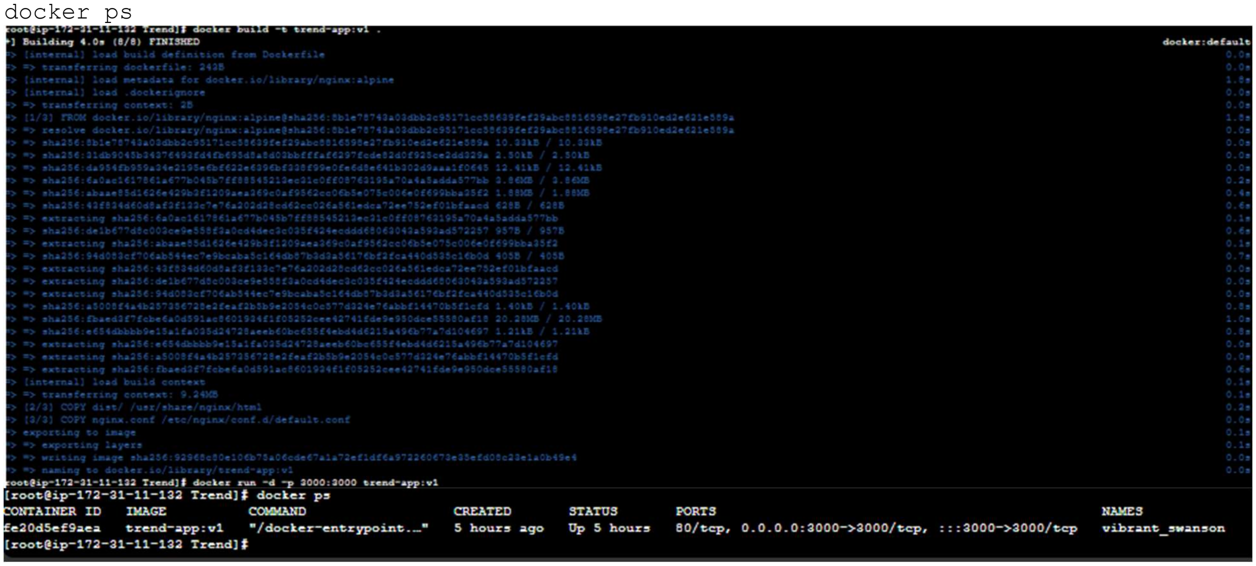
```
FROM nginx:alpine
COPY dist/ /usr/share/nginx/html
COPY nginx.conf /etc/nginx/conf.d/default.conf
EXPOSE 3000
CMD ["nginx", "-g", "daemon off;"]
```

3.2 nginx.conf

```
server {
  listen 3000;
  server_name _;
  root /usr/share/nginx/html;
  index index.html;
  location / {
    try_files $uri $uri/ /index.html;
  }
}
```

3.3 Build and Run

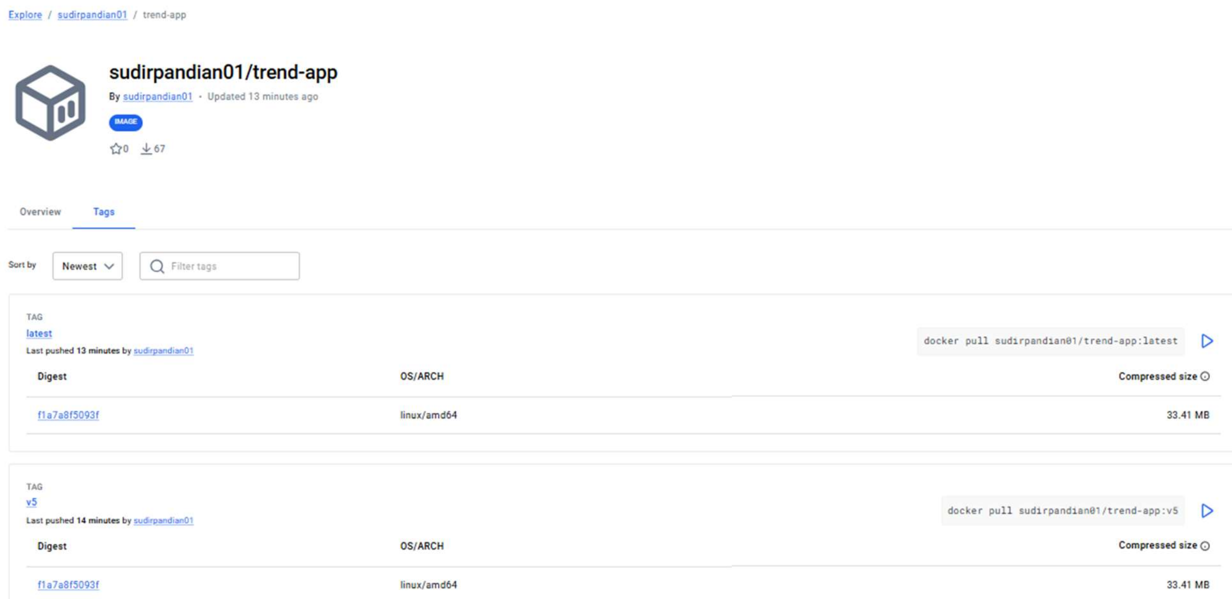
```
docker build -t trend-app:v1 .
docker run -d -p 3000:3000 trend-app:v1
docker ps
```



4. DockerHub

```
docker tag trend-app:v1 sudirpandian01/trend-app:v1
docker push sudirpandian01/trend-app:v1
```

DockerHub Repository: https://hub.docker.com/r/sudirpandian01/trend-app

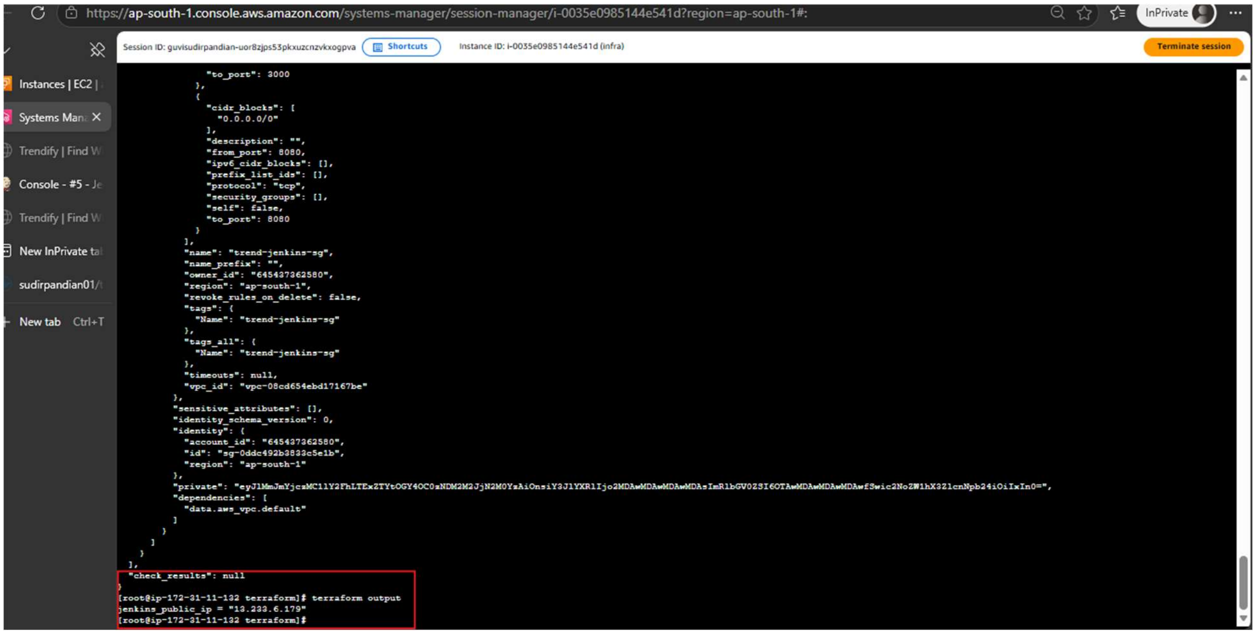


5. Terraform Infrastructure

Infrastructure defined in `terraform/main.tf` provisions the Jenkins server in the default AWS VPC.

Resource	Type	Details
VPC	Data source	vpc-08cd654ebd17167be (default)
Security Group	aws_security_group	Ports 22, 8080, 3000 open
IAM Profile	Data source	Existing profile: sudir
EC2 Instance	aws_instance	t3.medium, Ubuntu, Jenkins pre-installed

```
terraform init
terraform validate
terraform plan
terraform apply -auto-approve
```


https://ap-south-1.console.aws.amazon.com/systems-manager/session-manager/i-0035e0985144e541d?region=ap-south-1#:
Session ID: guvisudirpandian-uor8zjps53pkxuzcnzvkxogpva
Shortcuts
Instance ID: i-0035e0985144e541d (infra)
Terminate session
"to_port": 3000
"cidr_blocks": [
"0.0.0.0/0"
"description": "",
"from_port": 8080,
"ipv6_cidr_blocks": [],
"prefix_list_ids": [],
"protocol": "tcp",
"security_groups": [],
"self": false,
"to_port": 8080
"name": "trend-jenkins-sg",
"name_prefix": "",
"owner_id": "645437362580",
"region": "ap-south-1",
"revoke_rules_on_delete": false,
"tags": {
"Name": "trend-jenkins-sg"
"tags_all": {
"Name": "trend-jenkins-sg"
"timeouts": null,
"vpc_id": "vpc-08cd654ebd17167be"
"sensitive_attributes": [],
"identity_schema_version": 0,
"identity": {
"account_id": "645437362580",
"id": "sg-0ddc492b3833c5e1b",
"region": "ap-south-1"
"dependencies": [
"data.aws_vpc.default"
"check_results": null
[root@ip-172-31-11-132 terraform]# terraform output
jenkins_public_ip = "13.233.6.179"
[root@ip-172-31-11-132 terraform]#

6. Version Control

6.1 .gitignore

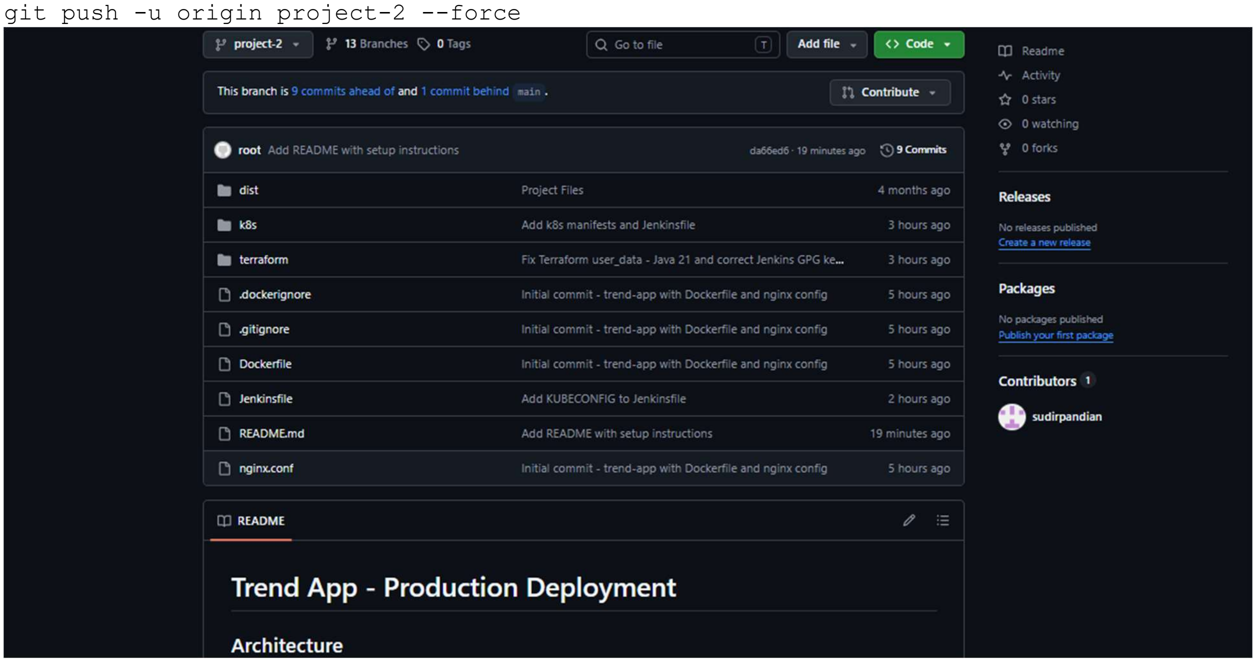
```
node_modules
.env
terraform.tfstate
terraform.tfstate.backup
.terraform/
*.pem
```

6.2 .dockerignore

```
node_modules
.git
.gitignore
README.md
*.pem
```

6.3 Push to GitHub

```
git init
git add .
git commit -m "Initial commit"
git branch -M project-2
git remote add origin https://github.com/sudirpandian/Devops.git
git push -u origin project-2 --force
```



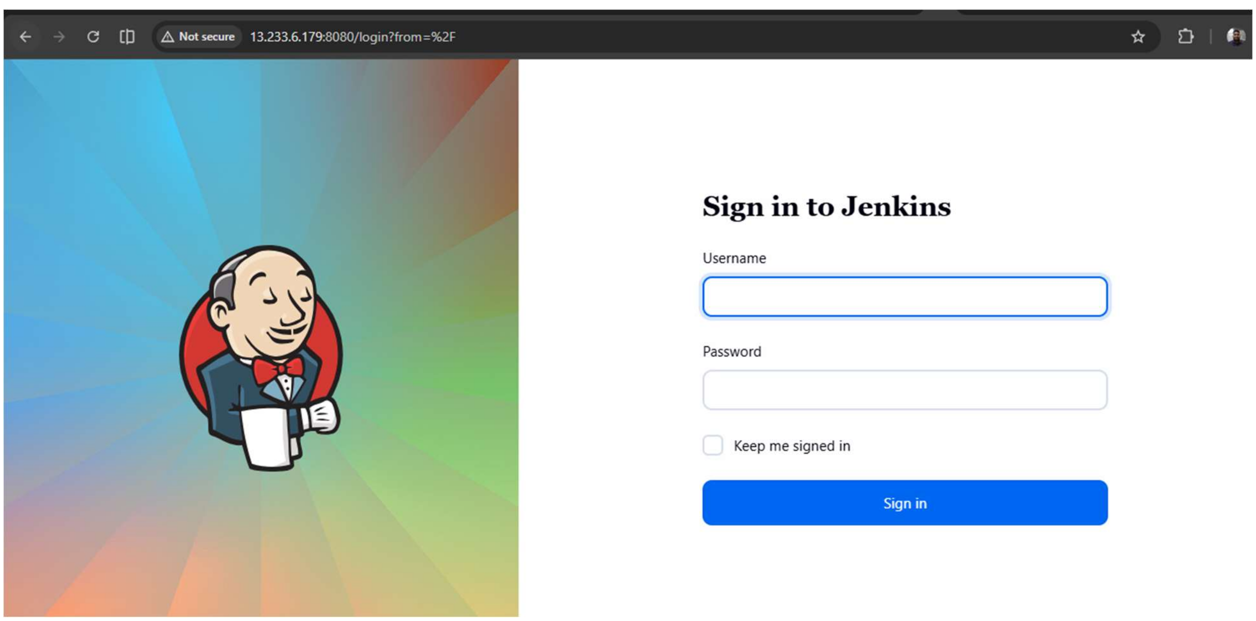
7. Jenkins

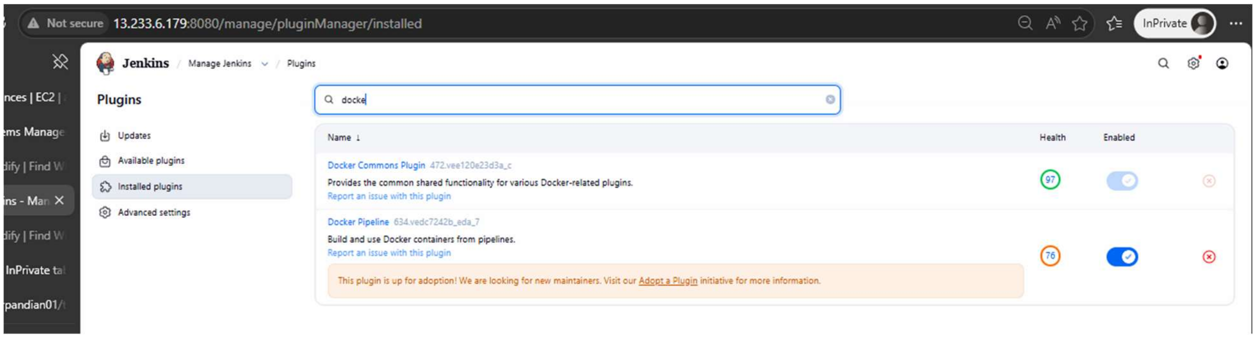
7.1 Installation

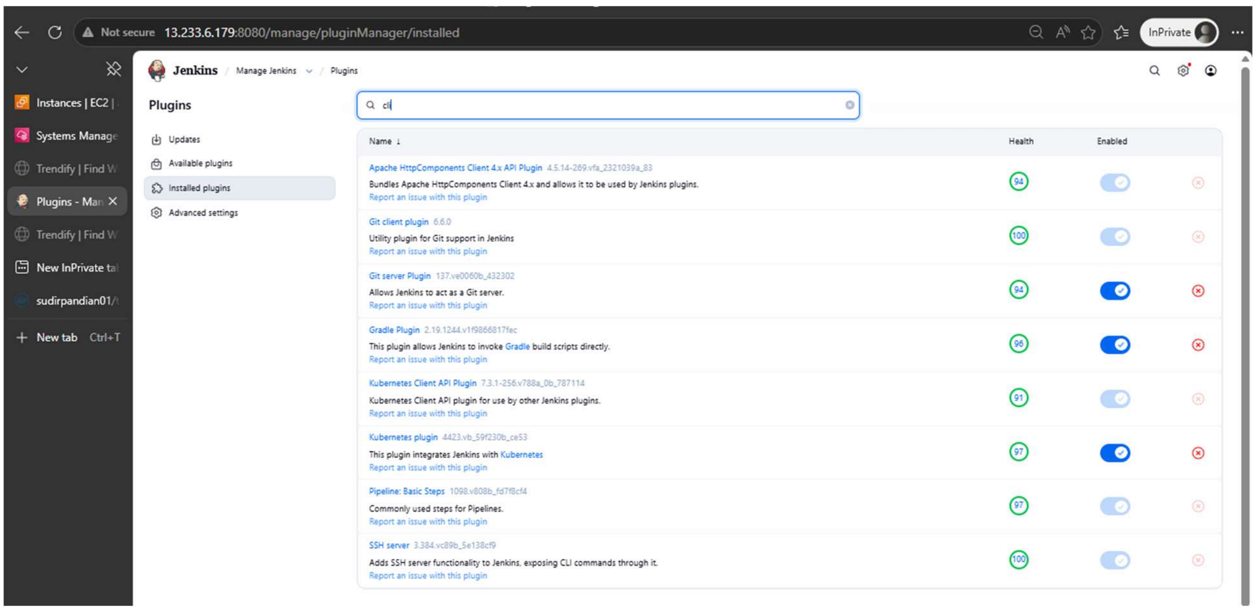
Jenkins is provisioned automatically via Terraform user_data on the EC2 instance at **13.233.6.179:8080**. Java 21 (OpenJDK) is required for Jenkins 2.555+.

7.2 Plugins Installed

Plugin	Purpose
Git	Source code checkout
Pipeline	Declarative pipeline support
Docker Pipeline	Docker build and push in pipeline
Kubernetes	Kubernetes integration
Kubernetes CLI	kubectl commands in pipeline
GitHub	GitHub integration
Credentials Binding	Secure credential injection



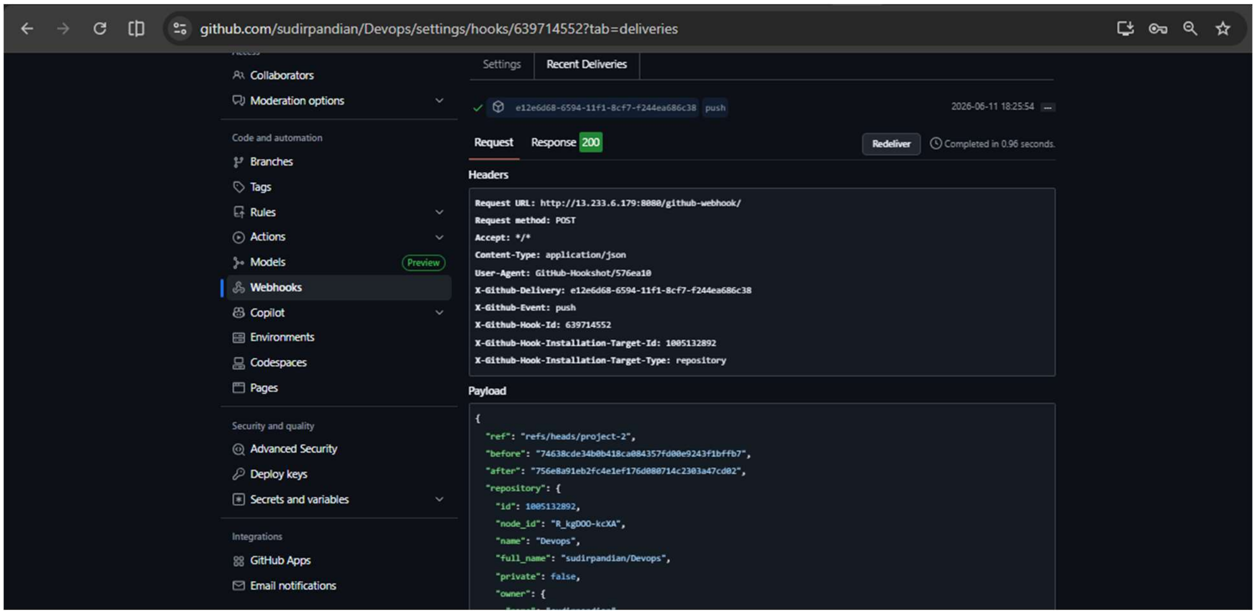




7.3 GitHub Webhook

Webhook configured at GitHub → Settings → Webhooks:

```
Payload URL: http://13.233.6.179:8080/github-webhook/
Content type: application/json
Trigger: Push events
```


github.com/sudirpandian/Devops/settings/hooks/639714552?tab=deliveries
Collaborators
Moderation options
Code and automation
Branches
Tags
Rules
Actions
Models
Preview
Webhooks
Copilot
Environments
Codespaces
Pages
Security and quality
Advanced Security
Deploy keys
Secrets and variables
Integrations
GitHub Apps
Email notifications
Settings
Recent Deliveries
e12e6d68-6594-11f1-8cf7-f244ea686c38
push
2026-06-11 18:25:54
Request
Response 200
Redeliver
Completed in 0.96 seconds.
Headers
Request URL: http://13.233.6.179:8080/github-webhook/
Request method: POST
Accept: */*
Content-Type: application/json
User-Agent: GitHub-Hookshot/576ea10
X-Github-Delivery: e12e6d68-6594-11f1-8cf7-f244ea686c38
X-Github-Event: push
X-Github-Hook-Id: 639714552
X-Github-Hook-Installation-Target-Id: 1005132892
X-Github-Hook-Installation-Target-Type: repository
Payload
"ref": "refs/heads/project-2",
"repository": {
"id": 1005132892,
"name": "Devops",
"full_name": "sudirpandian/Devops",
"private": false,
"owner": {

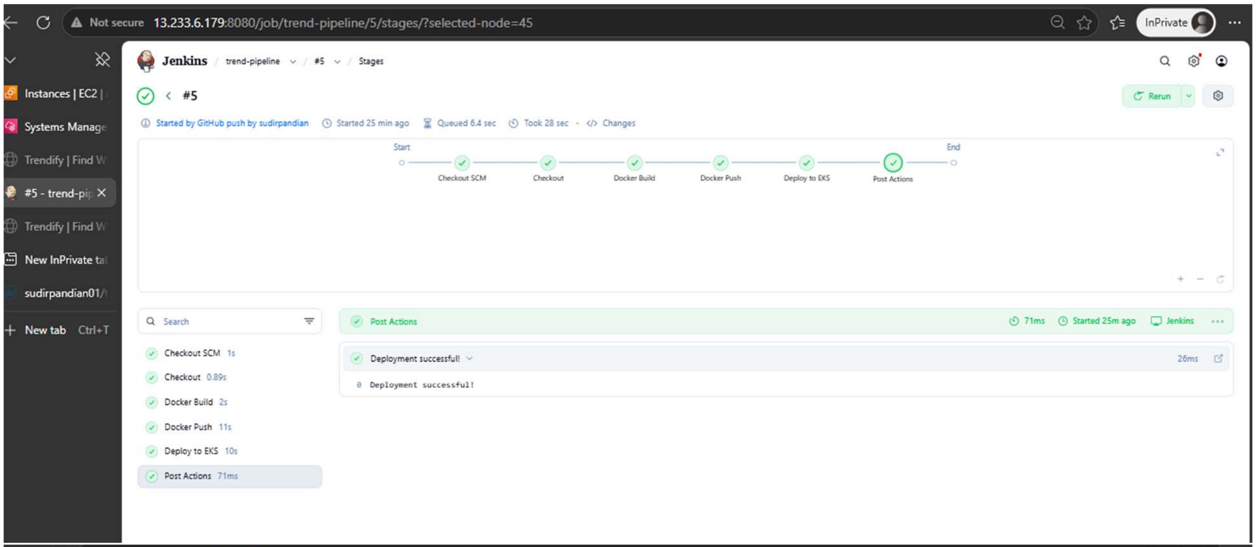
8. CI/CD Pipeline

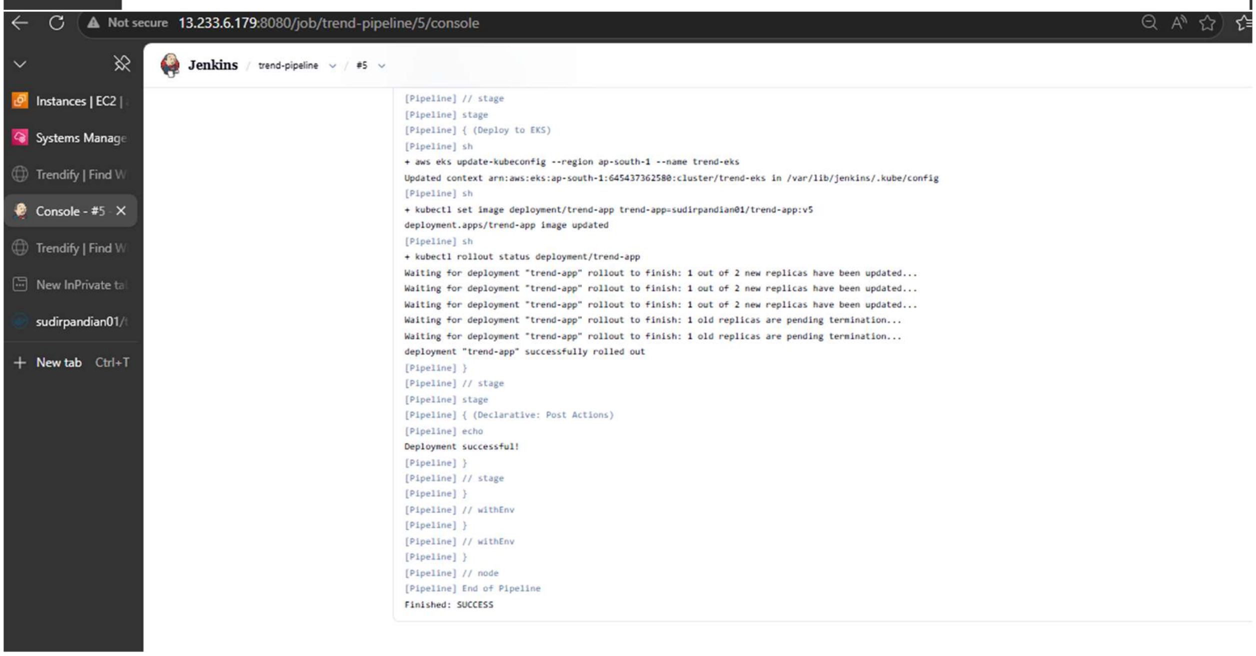
8.1 Pipeline Explanation

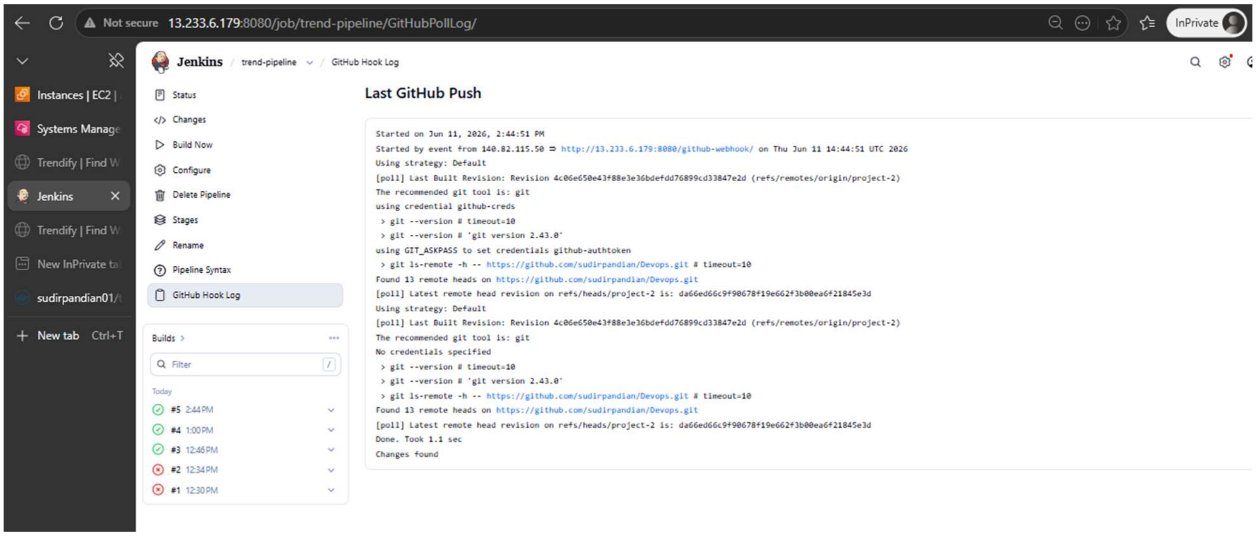
Stage	Action	Details
Checkout	Pull source code	Clones project-2 branch from GitHub
Docker Build	Build image	Tags with build number (v1, v2, v3...)
Docker Push	Push to DockerHub	Pushes versioned tag + latest
Deploy to EKS	Rolling update	kubectl set image + rollout status

8.2 Jenkinsfile

```
pipeline {
  agent any
  environment {
    DOCKERHUB_USER = 'sudirpandian01'
    IMAGE_NAME = 'trend-app'
    IMAGE_TAG = "v${BUILD_NUMBER}"
    EKS_CLUSTER = 'trend-eks'
    AWS_REGION = 'ap-south-1'
    KUBECONFIG = '/var/lib/jenkins/.kube/config'
  }
  stages {
    stage('Checkout') { ... }
    stage('Docker Build') { ... }
    stage('Docker Push') { ... }
    stage('Deploy to EKS') { ... }
  }
}
```


Not secure 13.233.6.179:8080/job/trend-pipeline/5/stages/?selected-node=45
Jenkins / trend-pipeline / #5 / Stages
#5
Rerun
Started by GitHub push by sudirpandian
Started 25 min ago
Queued 6.4 sec
Took 28 sec
Changes
Start
Checkout SCM
Checkout
Docker Build
Docker Push
Deploy to EKS
Post Actions
End
Search
Checkout SCM 1s
Checkout 0.89s
Docker Build 2s
Docker Push 11s
Deploy to EKS 10s
Post Actions 71ms
Post Actions
71ms
Started 25m ago
Jenkins
Deployment successful!
26ms
0 Deployment successful!

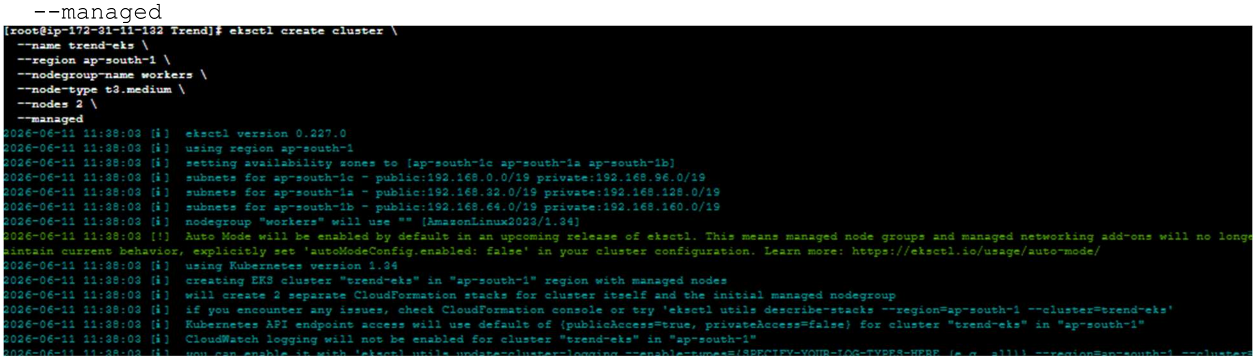

Not secure 13.233.6.179:8080/job/trend-pipeline/5/console
Jenkins / trend-pipeline / #5
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to EKS)
[Pipeline] sh
+ aws eks update-kubeconfig --region ap-south-1 --name trend-eks
Updated context arn:aws:eks:ap-south-1:645437362580:cluster/trend-eks in /var/lib/jenkins/.kube/config
[Pipeline] sh
+ kubectl set image deployment/trend-app trend-app=sudirpandian01/trend-app:v5
deployment.apps/trend-app image updated
[Pipeline] sh
+ kubectl rollout status deployment/trend-app
Waiting for deployment "trend-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "trend-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "trend-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "trend-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "trend-app" rollout to finish: 1 old replicas are pending termination...
deployment "trend-app" successfully rolled out
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Deployment successful!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS



9. Kubernetes — AWS EKS

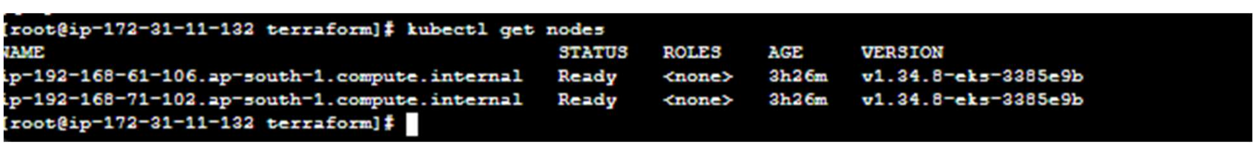
9.1 Cluster Creation

```
eksctl create cluster \
  --name trend-eks \
  --region ap-south-1 \
  --nodegroup-name workers \
  --node-type t3.medium \
  --nodes 2 \
  --managed
```



9.2 Verify Nodes

```
kubectl get nodes
```



9.3 Kubernetes Manifests

Files: `k8s/deployment.yaml` and `k8s/service.yaml`

```
kubectl apply -f k8s/
kubectl get pods
kubectl get svc
```

```
[root@ip-172-31-11-132 terraform]# kubectl get pods
NAME                         READY   STATUS    RESTARTS   AGE
trend-app-7b56c84f8-418lq    1/1     Running   0          29m
trend-app-7b56c84f8-t8k6w    1/1     Running   0          29m
[root@ip-172-31-11-132 terraform]#
```

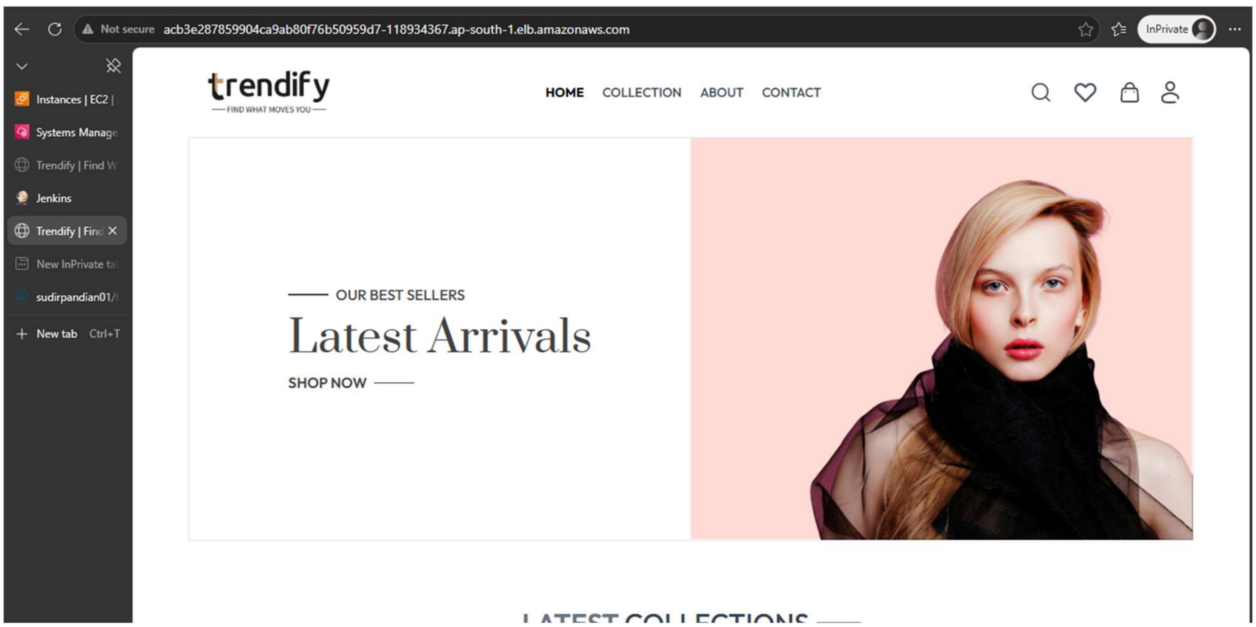
```
NAME            TYPE           CLUSTER-IP       EXTERNAL-IP                                                                PORT(S)        AGE
kubernetes      ClusterIP      10.100.0.1       <none>                                                                     443/TCP        3h32m
trend-service   LoadBalancer   10.100.164.171   acb3e287859904ca9ab80f76b50959d7-118934367.ap-south-1.elb.amazonaws.com   80:32485/TCP   3h19m
[root@ip-172-31-11-132 terraform]#
```

---

10. Application LoadBalancer

LoadBalancer DNS:
acb3e287859904ca9ab80f76b50959d7-118934367.ap-south-1.elb.amazonaws.com

Access URL: http://acb3e287859904ca9ab80f76b50959d7-118934367.ap-south-1.elb.amazonaws.com



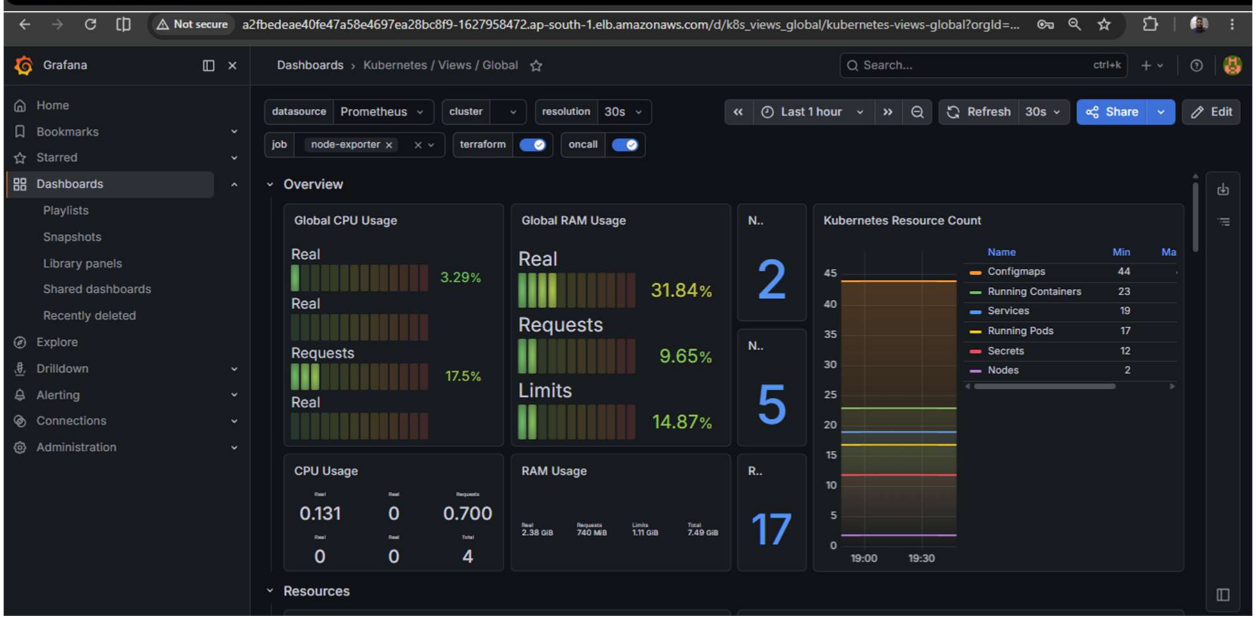
---

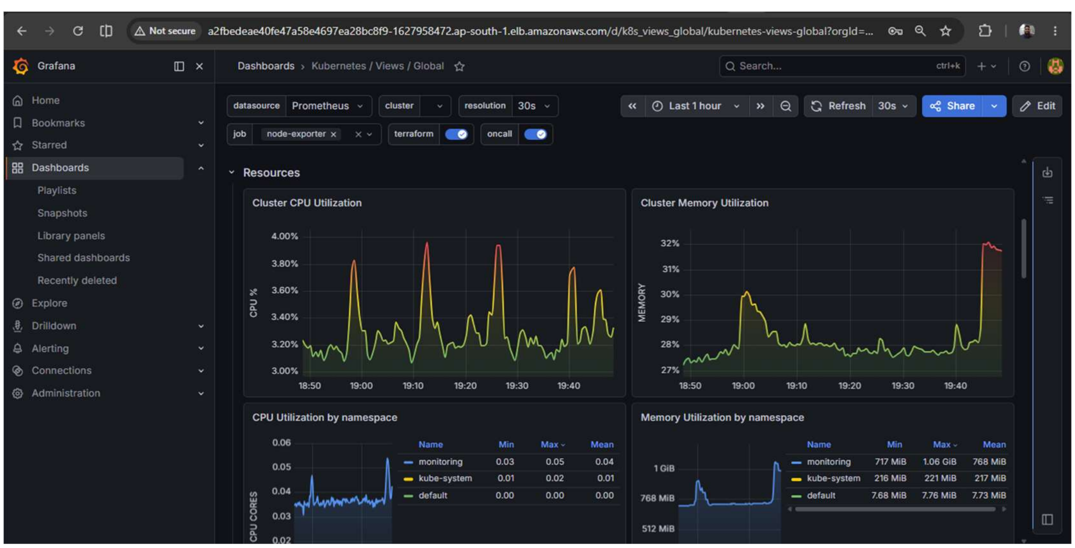
11. Monitoring

```
helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
helm install monitoring prometheus-community/kube-prometheus-stack \
  --namespace monitoring --create-namespace
kubectl get pods -n monitoring
```

Grafana URL: http://a2fbedeae40fe47a58e4697ea28bc8f9-1627958472.ap-south-1.elb.amazonaws.com
Login: admin / <secret from monitoring-grafana secret>

```
[root@ip-172-31-11-132 terraform]# kubectl get pods -n monitoring
NAME                                                     READY   STATUS    RESTARTS   AGE
alertmanager-monitoring-kube-prometheus-alertmanager-0   2/2     Running   0          147m
monitoring-grafana-5d4d9f9c49-wjwkn                      3/3     Running   0          147m
monitoring-kube-prometheus-operator-866bd47fdf-p6zg8     1/1     Running   0          147m
monitoring-kube-state-metrics-7f7f8474d9-gdchf           1/1     Running   0          147m
monitoring-prometheus-node-exporter-8vlqq                1/1     Running   0          147m
monitoring-prometheus-node-exporter-khh49                1/1     Running   0          147m
prometheus-monitoring-kube-prometheus-prometheus-0       2/2     Running   0          147m
[root@ip-172-31-11-132 terraform]#
```






CPU Utilization by instance
CPU %
5.00%
4.50%
4.00%
3.50%
3.00%
2.50%
19:00
19:30
Name Min Max
192.168.71.102:9100 3.54% 4.79%
192.168.61.106:9100 2.39% 3.31%
Memory Utilization by instance
MEMORY
1.75 GiB
1.50 GiB
1.25 GiB
1 GiB
768 MiB
19:00
19:30
Name Min Max
192.168.71.102:9100 1.32 GiB 1.68 GiB
192.168.61.106:9100 734 MiB 795 MiB
Kubernetes
Kubernetes Pods QoS classes
17.5
15
12.5
10
7.5
5
2.5
0
19:00
19:30
Name Min Max
Total pods 17 17
Burstable pods 9 9
BestEffort pods 8 8
Guaranteed pods 0 0
Kubernetes Pods Status Reason
100
80
60
40
20
0
19:00
19:30
Name Min Max
Evicted 0
NodeAffinity 0
NodeLost 0
PreemptionByScheduler 0
SchedulingGated 0
Shutdown 0
TerminationByKubelet 0
UnexpectedAdmissionError 0
Network
Global Network Utilization by device
BANDWIDTH
64 KiB/s
32 KiB/s
0 B/s
-32 KiB/s
-64 KiB/s
18:50
19:00
19:10
19:20
19:30
19:40
Network Saturation - Packets dropped
DROPPED PACKETS
100
80
60
40
20
0
18:50
19:00
19:10
19:20
19:30
19:40

12. Submission Summary

GitHub Repository	https://github.com/sudirpandian/Devops/tree/project-2
DockerHub Image	sudirpandian01/trend-app
Jenkins Server	http://13.233.6.179:8080
EKS Cluster	trend-eks (ap-south-1)
App LoadBalancer ARN	acb3e287859904ca9ab80f76b50959d7-118934367.ap-south-1.elb.amazonaws.com
Grafana	a2fbedeae40fe47a58e4697ea28bc8f9-1627958472.ap-south-1.elb.amazonaws.com